

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think> To solve this , we need to calculate the distance to his home using the total distance in the round trip .

- 1. First leg of his trip from his destination back home .
- 2. The stand still , and the low -speed drive will not affect the total distance until the resumed higher -speed trip .
- 3. Use the combined time , with their respective average final speeds for this leg .

John , to return from 3 hours at 60 mph ,
- Distance to return is $(3 \text{ hours} \times 60 \text{ mph} = 180 \text{ miles})$. So , the return distance without delay assumptions initially $(= 180 \text{ miles})$.

Next , treating the 2 hours stand still causing no added distance as described ;
- Next 30 minutes driven only covering distance from start home step back $(\text{ at } = 30 \text{ mph} \times 0.5 \text{ hours} = 15 \text{ miles})$.

After this phase resumes at 80 mph over last 1.5 hours ;

Total remaining Return traveled end phase . Base Phase $(= 180 - 15)$ from start .
So Re -calculated Dist returned average phase = Distance (Remain) * Speed then added upon .

Revised average target trip = $(165 - 15 = 165)$ in higher -speed final .

Verifying realistic achievable final Step where :

Breaks Logic).
Remains leg normal tripe only time left 2.5 final phase 80 for target at max return accumulation calculating .

Sum -Conjectures towards match ; simplified = Last's init steps ;

Thus (impact revision):

Final difference set : Total from end , fully remote check of ride final to home .

</think><answer> Thus John is <answer> 50 miles </answer> from home at the end of the 4 hours .</answer>